

PROPOSAL FOR

AUTOMATION OF WATER METER CALIBRATION TEST BENCH



Nagman Flow Level Systems and Solutions LLP

INTRODUCTION

ABOUT US

Nagman Flow Level Systems and Solutions LLP (www.nagmanflow.com) is a pioneer in providing cutting-edge flow and level measurement solutions. We specialize in developing water meter calibration test benches that ensure accuracy, reliability, and performance for industrial and utility applications. Our systems are designed to meet international standards and are used globally for testing and calibrating analog and digital water meters.

PROBLEM STATEMENT

The current manual process for testing and calibrating analog water meters is slow and prone to human error. This inefficiency reduces throughput, increases costs, and limits accuracy. Given the variety of water meter designs, a generic, scalable automation solution is required to streamline the calibration process, improve accuracy, and reduce errors.

PROJECT OBJECTIVE

The objective of this project is to develop an AI-based automated calibration system for water meter test benches. The system will:

- Capture Images: Automatically capture images of water meters using high-resolution cameras.
- Process Images with Machine Learning: Use advanced image processing and machine learning techniques to identify and interpret readings from analog meters.

- Handle Meter Variations: Implement a machine learning model to adapt to different water meter designs and layouts.
- Communicate Data: Send the extracted readings to a programmable logic controller (PLC) or external system for analysis and storage.
- Enable Integration: Ensure seamless integration with existing calibration workflows.

SCOPE OF THE PROJECT

HARDWARE

1. High-Resolution Camera Module:

- Capable of triggering image capture and transmitting data.
- Integration with PLC or external control.

2. Lighting and Mounting:

- Stable mounting setup for consistent image capture.
- Proper lighting setup to eliminate glare and shadows.

3. Communication Interface:

- Integration with PLC using GPIO, UART, or network interfaces (Wi-Fi/Ethernet).

SOFTWARE

1. Image Processing and Preprocessing:

- Use OpenCV for cropping, noise reduction, edge detection, and clarity enhancement.

2. Machine Learning (ML):

- Train and implement a machine learning model to identify and interpret various analog water meter designs.
- Use frameworks like TensorFlow or PyTorch for building the model.

3. OCR and Analog Dial Reading:

- Use OCR (e.g., Tesseract) to extract numerical readings.
- Implement algorithms to analyze needle positions and map them to meter values.

4. Data Handling and Communication:

- Transmit the results to the PLC or a database for calibration records.

APPLICATIONS

- Industrial Use: Streamlined and automated calibration of analog and digital water meters in testing and manufacturing facilities.
- Quality Control: Improved accuracy, reduced errors, and consistent readings for quality assurance.
- Skill Development: Provides students with practical exposure to AI, machine learning, and industrial automation.

STUDENT INTERNSHIP PROPOSAL

BENEFITS TO STUDENTS

Students will gain valuable real-world experience by working on an innovative and practical project. Key learning opportunities include:

1. Machine Learning: Develop and train ML models for object recognition and analog meter reading.
2. Computer Vision: Use OpenCV for image processing and analysis.
3. AI Edge Deployment: Learn to integrate AI solutions with edge devices for real-world applications.
4. Industrial Automation: Interface with PLCs and industrial systems.
5. Team Collaboration: Gain exposure to working in a professional engineering environment.

BASIC REQUIREMENTS FOR STUDENTS

- Knowledge of Python programming.
- Understanding of basic machine learning concepts.
- Familiarity with image processing tools (e.g., OpenCV).

RECOMMENDED REQUIREMENTS

- Prior experience with TensorFlow, PyTorch, or similar ML frameworks.
- Understanding of embedded systems and industrial protocols.

- Familiarity with hardware interfacing (e.g., GPIO, UART).
- Strong analytical and problem-solving skills.

HOW TO APPLY

Interested students can sign up for the internship program by providing their:

- Name
- Academic Background
- Skills/Experience
- Contact Information

Next Steps: Shortlisted candidates will be invited for an interview to assess their skills and interest in the project. We would prefer a group of 2-3 candidates who can showcase their skill and express their desire to collaborate with us on this project.

Note: This is a hands-on project that requires students to collaborate closely with our team on-site at our premises. Remote participation is not an option; students must be physically present to fully engage in project activities and collaborate effectively

CONCLUSION

At Nagman Flow Level Systems and Solutions LLP, we are committed to driving innovation in industrial automation and calibration. This project will revolutionize the water meter calibration process while providing engineering students with hands-on experience in cutting-edge AI and machine learning technologies. We look forward to partnering with your institution and nurturing the next generation of engineers.

CONTACT INFORMATION

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